Recent Progress in Palaeomagnetism in Japan

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The scientific studies of determining the direction of geomagnetic field in the old geologic times from the data of the remanent magnetization of rocks have been continued by us since the pioneer work carried out by M. Matuyama in 1929.(1)

The specimens examined in our studies are the lava flows and the horizontal sedimentary layers of various localities of Japan. The rather intense magnetization of volcanic rocks were measured by astatic magnetometers both in Tokyo and Kyoto, while the weak magnetization of specimens of sedimentary layers were examined by the induction-type equipment(2) or by a magnetometer of special type(3) in Tokyo and by a resonance-type magnetometer newly designed(4) in Kyoto.

From each lava flow, specimen pieces of five to twenty in number were cut off, the direction of the pice *in situ* being measured. On the other hand, a large number of specimens were systematically taken in various different localities and depths of each layer.

The final results of measurements are summarized in Tables I and II, where the mean values of dip and the deviation of declination from its present value are shown together with the localities, petrological characters and ages of rocks.

As to the reliability of direction of remanent magnetization of volcanic lavas, the result shown in the upper part of Table II will give a significant proof. The ejection and solidification of these lavas were actually observed, and it was firmly ascertained that their thermo-remanent magnetism keeps the direction of geomagnetic field in their neighbourhood. Thus we may say that, so far as we select the large and unbroken lava, the direction of their remanent magnetization can indicate that of geomagnetic field at the time of their cooling.

As will be seen in Table I, the volcanic lavas and sedimentary layers ejected and deposited respectively during the geological times from Upper Neogene through Lower Holocene almost Keep the direction of their magnetisation nearly equal to the present value of geomagnetic force, except the Azuki-tuff (Upper Neogene deposits), the lava at volcano Sigi (Upper Neogene ejecta) and the dyke rock at Kasuga (Lower Holocene).

Table I. Geological Ages.

No.	Locality	Rock	Age	ΔD(E)	I	Observer
1	Kasuga (Nara)	Dolerite dyke	Lower Holocene	76°0±0°4	30°5±2°4	Kawai
2	Mikasa (Nara)	Two-pyroxene andesite	Lower Holocene	−1.1±1.8	36.9±3.8	Kawai
3	Kabuto (Hyōgo)	Bronzite andesite	Lower Holocene	−8.4±2.2	49.8±1.8	Kawai
4	Tiba	Narita-bed (fine sand layer)	Pleistocene	4.5	56.3	Nagata*
5	Yawatano (Amagi)	Two-pyroxene andesite	Pleistocene	12.7±1.5	47.3±1.6	Nagata
6	Omuro	Olivine-basalt	Pleistocene	8.3±4.2	44.3±3.8	Nagata
7	Aziro	Olivine-augite basalt	Lower part of Pleistocene	−26.7±2.0	49.7±1.5	Nagata
8	Osaka	Azuki-tuff	Upper part of upper Neogene	−159.5±1.8	−126.3±6.3	Kawai
9	Sigi (Osaka, Nara)	Two-pyroxene andesite	Upper part of upper Neogene	−168.4±4.6	−136.8±10.8	Kawai
10	Odake (Nara)	Sanukitic andesite	Lower part of upper Neogene	−13.5±1.7	25.5±6.3	Kawai
11	Nizyo 1. (Nara)	Bronzite andesite	Lower part of upper Neogene	9.3±2.6	54.3±1.9	Kawai
12	Nizyo 2. (Nara)	Biotite-hornblende andesite	Lower part of upper Neogene	−8.8±5.3	60.6±6.9	Kawai
13	Myōzon (Nara)	Hornblend Hypersthen andesite	Lower part of upper Neogene	−8.9±2.4	54.3±2.2	Kawai
14	Unebi (Nara)	Biotite andesite	Lower part of upper Neogene	−9.8±1.9	45.3±2.1	Kawai
15	Miminasi (Nara)	Biotite andesite	Lower part of upper Neogene	−7.0±3.4	51.2±2.7	Kawai
16	Sidara (Aiti)	Basaltic andesite	Miocene	28.3±2.9	65.7±8.8	Sannodo & Magata

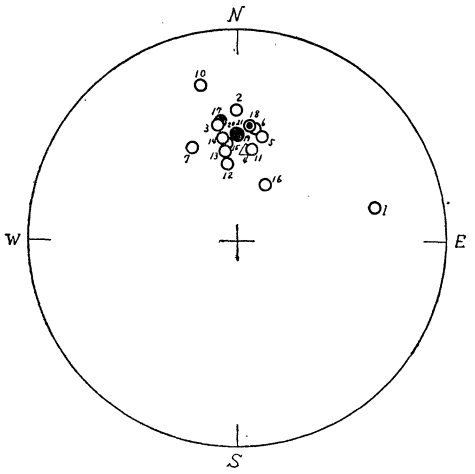
* The average value of 271 samples collected from the whole parts of layer of 7 m in thickness.

Table II. Present and Historical Ages.

No.	Locality	Rock	Age	ΔD(E)	I	Observer
17	Mihara (Oosima)	Basalt	1914	−0°3±0°7	49°0±0°9	Nagata
18	Mihara (Oosima)	Basalt	1940	−0.0±0.7	48.4±0.6	Nagata
19	Miyake-zima	Olivine-pyroxene basalt	1940	−0.1±0.5	48.2±0.4	Nagata
20	Aokigahara (Huzi)	Basalt	864	−7.9±0.9	43.3±0.6	Nagata
21	An'ei lava (Mihara)	Basalt	1778	6.4±1.9	44.3±1.6	Nagata

Fig. 1 a.

Lower hemisphere

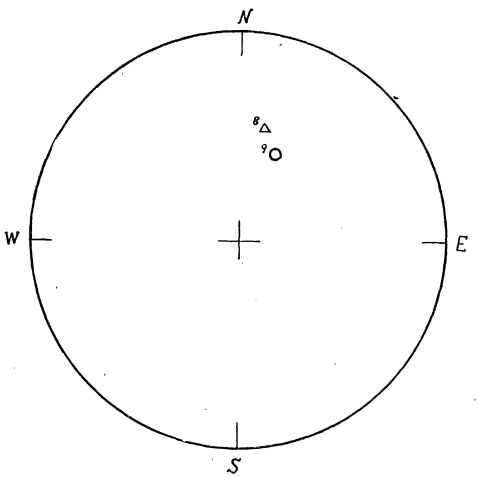


North seeking polarizations

△ Sedimentary rock
○ Geological age volcanic rocks
◉ Historical age volcanic rocks
● Present volcanic rocks

Fig. 1 b.

Lower hemisphere



South seeking polarizations

△ Sedimentary rock
○ Volcanic rock

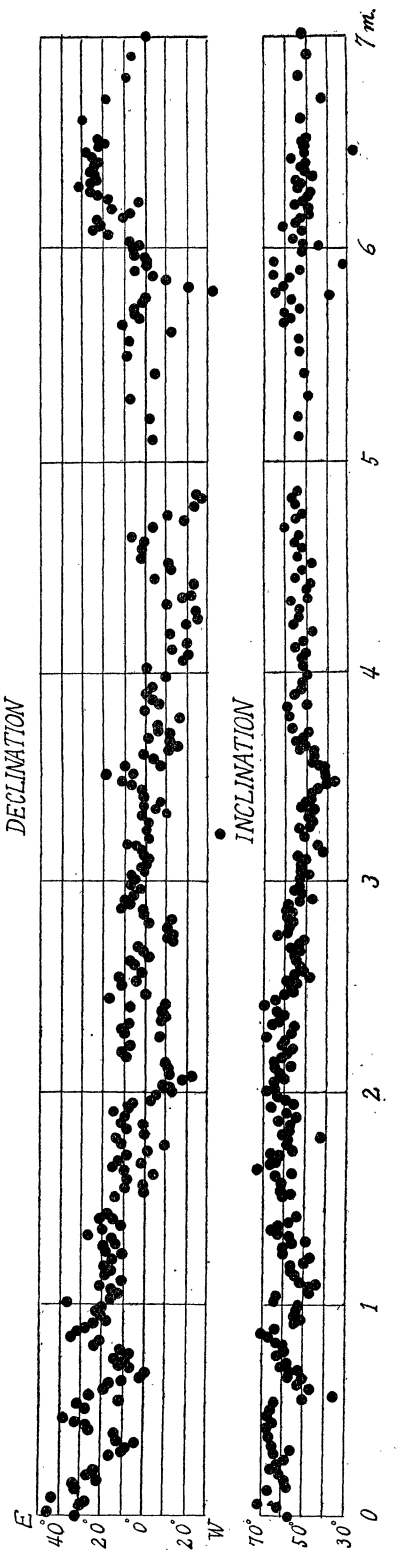


Fig. 2. Changes in Direction of Remanent Magnetization of Narita-bed with Depth.

The specimens collected from various localities in the Azuki-tuff layer around Osaka city are reversely magnetized, while the lava specimens from volcano Sigi, one of the Upper Neogene lava flows situated around the tuff layer are reversely magnetized, though the others are almost normally magnetized.

On the other hand, the sedimentary layers in Bōsō peninsula near Tokyo throughout the ages from Miocene to Pleistocene have the direction of magnetization nearly equal to the present geomagnetic force. Fig. 2 shows the direction of horizontal component of remanent magnetization of these layers, while Fig. 3 shows the change with depth in the direction of magnetization of the newest layer among them.(5)

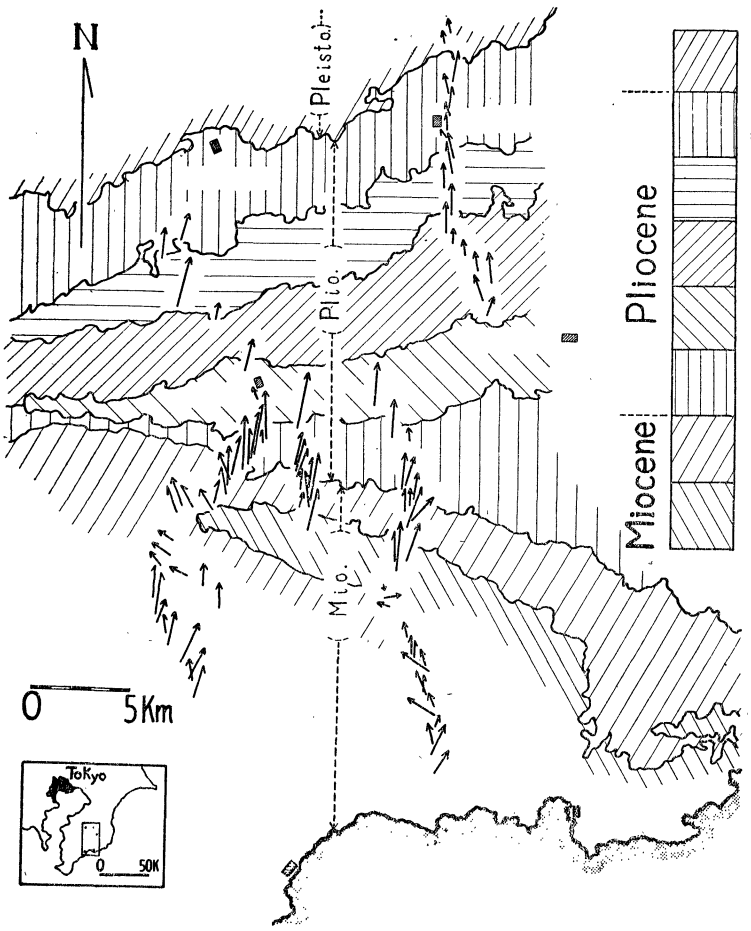


Fig. 3. Magnetization of Tertiary Sediments in Bôsô Peninsula, Japan.

The causation of the reverse magnetization of the Azuki-tuff and of volcano Sigi has not yet been clarified. It can be said, however, that the reverse magnetization of rock masses can be found only in Tertiary ejecta and deposits, and hardly found in Quaternary ejecta and deposits, so far concerning Japanese data.

It will be worthwhile to note that the data of reverse magnetization hitherto

found in England[6] and in Japan[1] are concerning the Tertiary ejecta and intrusions respectively.

In order to promote palaeomagnetism in future, the comparison of the similar data obtained over the world will be most fundamental. For this purpose, the identification of geological ages will become the most significant element.

Especially, the international cooperation in the systematic studies of horizontal strata of sediments such as carried out by McNish, Johnson, Murphy, Torreson, Graham[7, 8, 9] and us will be much desirable, since geological identification of sediments seems to be more reliable.

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